Course Name: Midterm Exam Sheet

Open-book reference

Student: Your Name (YOUR-ID)

Exam playbook (read this first)

Exam conditions. Open book? Printed material only? Calculator? Solver? (If no solver: every answer below is done by hand.)

The announced questions (copy the teacher's message word for word):

Q	What she will ask	Kind	Go to
Q1	e.g. formulate an LP, its dual, complementary slackness	computation	Playbook Q1, p. 2
Q2	e.g. prove a function / problem is convex	**proof**	Playbook Q2, p. 3

Habits that save marks.

- Name the method before using it ("by the second-order condition …").
- Write the general formula first, then plug in the numbers.
- Check the answer: constraints hold, multipliers have the right sign, primal value = dual value.
- For proofs: state the claim, the method, each step with its reason, then □.

Hand-calculation helpers. Small formulas you keep looking up.

Playbook Q1: name of the question type

Computation. Lectures N–M.

Q1 What the teacher announced — Announcement, <date>

Copy the exact wording of this announced question.

What a full-mark answer contains. A short checklist of the parts the marker looks for.

Step-by-step guideline — works for any numbers

STEP 1 **First thing to write**

Written with letters, so it works for any problem.

STEP 2 **Next thing**

...

STEP 3 **Check**

How to verify the answer by hand.

Worked example

> A typical problem of this type (from the review sheet or a past paper).

STEP 1 **First thing to write**

The real numbers, every line.

STEP 2 **Next thing**

...

Answer: Q1 (example) — what to write on the exam paper

(a) Short working, final answer boxed: $\boxed{x^\star = (\dots)}$

Traps

The mistake most people make on this type, and a quick check that catches it.

Playbook Q2: name of the proof question

Proof. Lectures N–M.

Q2 What the teacher announced — Announcement, <date>

e.g. "Prove that the problem … is a convex (concave) optimization problem."

Which method to pick.

If you see …	use …
a twice-differentiable function	second-order condition: Hessian $\succeq 0$
a sum / max / composition of known pieces	the rules that keep convexity

Step-by-step guideline — works for any numbers

STEP 1 **State the claim**

What exactly is being proved, and on which domain.

STEP 2 **Pick and name the method**

…

STEP 3 **Prove each piece**

One line per piece, with its reason.

STEP 4 **Conclude**

Tie the pieces together and finish with □.

Proof: template for this question type — copy this structure

Claim. …
Method. …
Step 1. … *(reason)*
Step 2. … *(reason)*
Conclusion. Hence … is convex. □

Worked example

Proof: a real example, filled in — copy this structure

Claim. $f(x) = x \log x$ is convex on $x > 0$.
Method. Second-order condition.
Step 1. $f'(x) = \log x + 1$, $f''(x) = 1/x$.
Step 2. For $x > 0$, $f''(x) = 1/x > 0$ *(a positive number over a positive number)*.
Conclusion. $f'' > 0$ on the convex domain $x > 0$, so f is strictly convex. □

Practice questions

Review sheet / exercises in their original order. Each says which exam question it trains.

Question 1: short name (trains Q1)

Question 1 Review sheet, p. 1

Copy the question exactly as printed.

Step-by-step guideline works for any numbers

STEP 1 **Same recipe as Playbook Q1**

Point back to it, and note anything different.

STEP 1 **...**

Worked with the real numbers.

Answer: Question 1 what to write on the exam paper

(a) ...

Theory reference

By lecture. Each section says which exam question it feeds.

Symbol dictionary (read this first)

Symbol	Read it as …
$x^\star$	the optimal point
∇f	the gradient: the list of slopes

Topic name (Exam Q1, Lecture 1)

> **Lecturer's definition** (Lecture 1, slide 5)
> The lecturer's exact definition. If the slide has a list of N items, write all N.

In plain words: one or two sentences. Then a small example with numbers.

Appendix

Tables to look things up fast.

Derivative and formula tables

$f(x)$	$f'(x)$	$f(x)$	$\nabla f(x)$
x^n	nx^{n-1}	$a^T x$	a
$\ln x$	$1/x$	$x^T Ax$	$(A + A^T)x$
e^{ax}	ae^{ax}	$\|x\|_2^2$	$2x$

Slide index

Topic $\rightarrow$ lecture and slide number, so you can find the original fast during the exam.